

Enhancing the Human Tactile Sensitivity through an Individually Optimized Imperceptible Vibration Stimulation

Takashi Suzuki

takashi.suzuki.fj@kyocera.jp

Kyocera Corporation

Yokohama, Kanagawa, Japan

ABSTRACT

Stochastic Resonance (SR) is known as a method to increase tactile sensitivity by applying weak vibrations. We investigated how the tactile sensitivity can be enhanced using sinusoidal vibrations with a specific frequency instead of conventionally applied white noise. The results demonstrated that sensory sensitivity with sinusoidal vibrations between 200 and 400 Hz increased much more than that obtained with white noise vibrations. To confirm the enhanced sensitivity effects, we propose a test based on cloth discrimination. Accordingly, it was found that the average correct rate increased using vibrations generated with optimal frequency compared to white noise vibrations. Notably, some participants stated a significant increase in tactile sensitivity when using the optimal vibration for SR. As a result, we showed that by using the optimal sinusoidal wave noise for each participant, it is possible to improve the tactile sensitivity far beyond that achieved with conventional SR using white noise.

CCS CONCEPTS

• **Hardware** → **Haptic devices**; • **Human-centered computing** → *Laboratory experiments*.

KEYWORDS

stochastic resonance, tactile, tactile sensitivity, vibration, white noise, sinusoidal vibration, fingertip, cloth discrimination

ACM Reference Format:

Takashi Suzuki. 2024. Enhancing the Human Tactile Sensitivity through an Individually Optimized Imperceptible Vibration Stimulation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3650826>

1 INTRODUCTION AND PREVIOUS RELATED WORKS

Enhanced tactile sensitivity in the fingers improves a person's ability to sense the texture and shape of objects in detail, enhances object recognition, and improves work performance and safety. Various methods have been proposed to improve tactile sensitivity, such as attaching a sensor to the finger to amplify the tactile

sensation and presenting it at another location known as tactile transfer. To directly augment human sensation, we focus on ways to improve the sensitivity of human fingertips themselves. A common approach to improve tactile sensitivity is using the stochastic resonance (SR) effect, which increases sensitivity by introducing appropriate noise into nonlinear systems, such as the nervous system [1, 2, 5, 6, 8, 9, 11–16, 19, 20, 22]. However, the SR effect has yielded limited improvements in tactile enhancement reported so far [1, 2, 9, 11, 13, 14, 16, 19, 22], and it is important to improve the SR effect performance compared with that of conventional methods.

Traditionally, white noise has been used to generate the SR effect [1–22]. However, we demonstrated that using sinusoidal vibrations was superior to using white noise regarding SR-effect induced improvements in the tactile sensitivity of fingers. The main contributions of this study are twofold: 1) We demonstrated and evaluated an original method using the SR effect to achieve greater tactile enhancement compared with that achieved using conventional methods. 2) We proposed a practical task and validated our method to enhance tactile sensitivity.

2 ENHANCEMENT OF FINGERTIP TACTILE SENSITIVITY BY OPTIMAL VIBRATION

In this section, we describe how the sensory threshold of the fingertips is measured as weak vibrations that cause SR using white noise and sinusoidal vibrations at various frequencies. We also present the differences among participants regarding the type of noise that optimizes the sensory expansion.

2.1 Participants

There were 11 participants (three women, eight men; company employees who were not involved in the project) with a wide age range of 23–60 years (average=39.1, SD=13.2). Participation was voluntary, and the ethical guidelines of the company and those established according to the Declaration of Helsinki for research involving Human Participants were adhered to. All participants were healthy and free from detectable neurological disorders.

2.2 Measurement method and equipment

A previously reported experimental method was used to confirm the SR effect [11, 16]. When an imperceptible vibration is applied to a finger, the change in the sensory threshold of the fingertip in response to the white noise vibrations is measured. Because the noise source should be sufficiently small to be worn on a finger and requires uniform vibration characteristics over a wide frequency range, a small voice coil-type vibration actuator (No. 602760, Foster Electric Company Limited, Japan) was adopted in this study.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650826>

Figure 1 shows a schematic of the experimental equipment. To avoid interference from the fingertip, a vibration actuator was attached between the first and second joints of the non-dominant index finger using elastic tape. Because all participants were right-handed, the vibration actuator was attached to the index finger of the left hand during the experiments. The vibration source signal was created using a PC (PC1) software. A white noise signal was obtained by applying a low-pass filter with a cutoff frequency of 500 Hz to the signal generated using the Box-Muller method, considering the sensitivity of the receptors. The signal was amplified using an amplifier (PAM8012, Diodes Incorporated, USA) to generate the vibrator input signal.

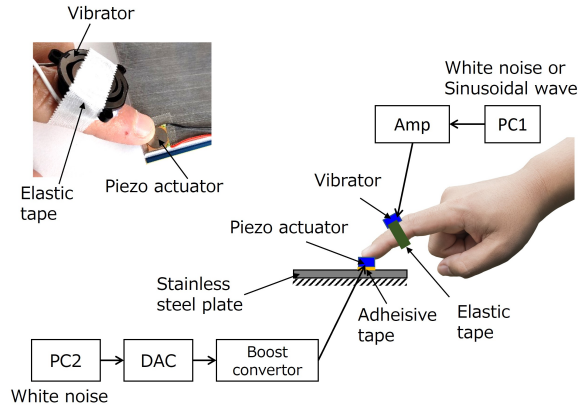


Figure 1: Schematic of the experimental setup used for testing the SR effect. The piezo actuator on the SUS board is operated by the white noise generated using the PC (PC2) after passing through the digital-analog converter (DAC) and boost convertor. The vibrator, which is worn on the finger, operates using a signal created by another PC (PC1), which is fed through an amplifier. For this signal, either a 100–400 Hz sine wave or white noise was select randomly. The presented picture shows the experimental view.

To determine the sensory threshold for vibrations with and without the SR effect, the participants were prompted to touch another fixed vibrator with the tip of the index finger to which the vibrator was attached. As shown in Figure 1, a piezo actuator (0909H011V060, TDK Electronics AG) was fixed to the surface corner of a stainless-steel plate (50mm×50mm×5mm) with an adhesive tape. To drive this piezo actuator, another PC (PC2) was used to generate an uncorrelated signal using the corresponding software via the Box-Muller method. The signal passed through a low-pass filter with a cutoff frequency of 500 Hz and was converted into an analog signal using a digital-analog amplifier (PC100USB-HR2, FOSTEX CO., Foster Electric Company Limited, Japan). The input signal for the piezo actuator was generated by amplifying the analog signal using a boost converter (DRV2667, Texas Instruments, Inc.).

2.3 Procedure

A schematic of the experimental procedure is shown in Figure 2, which consists of the following three steps as a single unit.

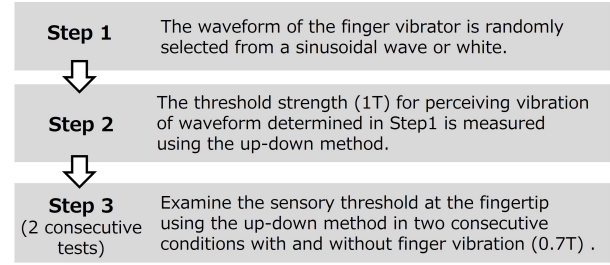


Figure 2: Outline of experimental procedure implemented for investigating the SR effect. One unit consists of three steps. First, the excitation waveform is decided in step 1. Then, the threshold value for the finger vibration is measured in step 2. Finally, in step 3, the sensitivity of the fingertip is determined with and without excitation vibration.

Step 1: The waveform of the finger vibrator (PC1) was randomly selected from white noise or sinusoidal waves with frequencies of 100, 200, 250, 300, or 400 Hz.

Step 2: The participant held the index finger of their non-dominant hand in the same relaxed position as when tracing the vibrator on the stainless-steel plate. The threshold intensity (1T) for perceiving vibration was determined by incrementally increasing or decreasing the finger's vibration intensity repeatedly until the participant was barely able to distinguish between the vibration source being turned on and off.

Step 3: The participant traced the piezo vibrator with the tip of their index finger, and the threshold intensity for perceiving the vibration was measured under two consecutive conditions, similar to that reported in Step 2. One condition was without vibration, while the other condition was with vibration intensity of 0.7T based on previous studies. The two conditions were adopted randomly. Because the vibrator attached to the participant's finger did not produce any perceptible sound during the operation, the participant could not perceive the difference between the two measurement procedures.

2.4 Statistical analysis of data

As two units were conducted per waveform, 12 tests (six waveforms×2) were conducted for each participant. To analyze the results, we calculated the relative change in the sensory threshold under vibration conditions for each test unit, using the condition without vibration as a reference [16]. We performed a two-tailed paired t-test to confirm the differences between cases with and without vibration.

2.5 Results and discussion

The results are shown in Figure 3. The horizontal axis indicates the vibration conditions, whereas the vertical axis shows the average sensory threshold change compared to the no-vibration condition for each participant. This figure shows that the sensory threshold of the fingertip decreases, while the tactile sensitivity increases using sinusoidal waves rather those observed when using white noise ($t=1.816$, $DOF=21$, $p=0.084$). These differences are statistically significant at 100, 200, 250 and 300Hz.

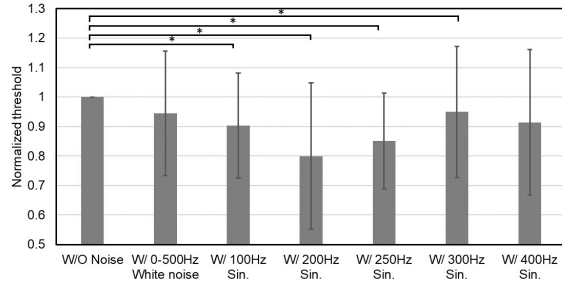


Figure 3: Changes in fingertip tactile sensitivity when using various excitation waveforms. The vertical axis shows the relative change in tactile sensitivity of the fingertip with respect to the condition without vibration. * indicates a significant difference ($p < 0.05$).

However, the frequency at which the tactile sensation peaks varies, and Figure 4 shows the optimal noise frequency distribution that best improves the tactile sensitivity for each participant. Only one participant (#1) showed the highest tactile sensitivity when using white noise rather than sinusoidal vibrations, but the frequency with the highest tactile sensitivity within sinusoidal vibrations is plotted. The optimal frequency is most often 200 Hz; however, it ranges between 200–400 Hz. Subsequently, we investigated the level at which the tactile sensation was the most sensitive, assuming that the optimal frequency was applied to each participant, as shown in Figure 5. This study revealed that if the noise waveforms were optimized for each participant, tactile sensitivity could be improved by 28% on average. Because the improvement effect of white noise was only 6% in Figure 5 and the previous studies conducted using similar measurement methods while implementing white noise have improved the tactile sensitivity by approximately 15% [16], we state that the tactile sensitivity can be significantly improved compared with the previous results by implementing the optimal sinusoidal frequency.

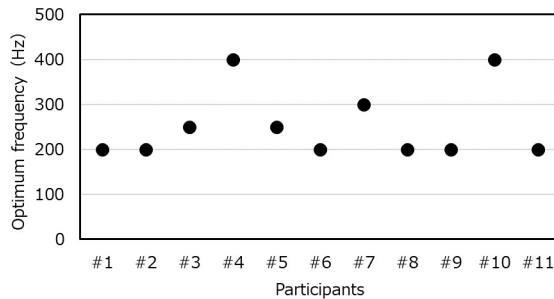


Figure 4: Distribution of frequencies that maximize the SR effect for each participant. The optimal frequency ranges between 200–400Hz; 200 Hz was the most commonly preferred optimal frequency.

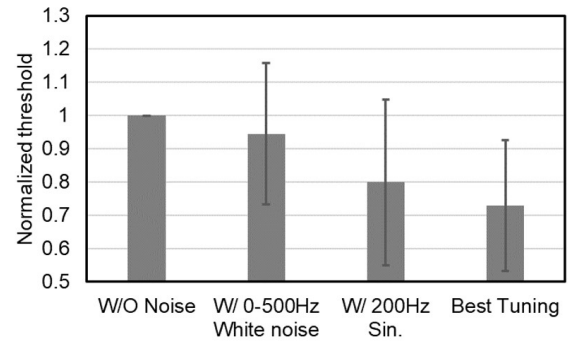


Figure 5: Changes in fingertip tactile sensitivity observed under various excitation waveforms. The vertical axis denotes the relative change in tactile sensitivity of the fingertip with respect to the condition without vibration. "Best tuning" denotes the results obtained assuming that each participant used optimal sinusoidal frequency at which their tactile sensitivity increased the most.

3 VERIFICATION OF OPTIMAL VIBRATION EFFECT THROUGH CLOTH DISCRIMINATION TEST

In this Section, we demonstrate the effects of noise optimization for each participant using the SR in practical applications. The methods for confirming the SR effect reported so far have certain drawbacks, namely, being impractical or damaging the fingertips [9, 11, 14, 19]. In this study, we devised an experimental plan based on cloth discrimination without damaging the fingertips and investigated the effect of changing the tactile sensitivity depending on the vibration types reported in Section 2.

3.1 Participants

We randomly selected 7 participants from among those who participated in the previous experiment, and one participant who did not participate in the previous experiment (1 woman, 7 men: company employees who were not involved in the project) over a wide age range of 28–60 years (average=44.1, SD=10.5). Participation was voluntary, and the ethical guidelines of the company and those established according to the Declaration of Helsinki for research involving Human Participants were adhered to. All participants were healthy and free from detectable neurological disorders.

3.2 Measurement equipment and procedures

As described in Section 2, a vibrator was attached between the first and second joints of the index finger of the non-dominant hand as a noise source. The vibrator control setup and measurement procedures regarding each participant's tactile threshold for white noise and sinusoidal noise were the same as those described in Section 2. The vibration intensity was set to 0.7T for each participant with respect to each participant's optimal frequency for SR or white noise. The seven participants, who also participated in the experiment described in Section 2, used the individual optimal

frequency found at the time, while the remaining person, who did not participate the previous experiment, used a 200Hz vibration.

To confirm the differences observed in the tactile sensation depending on the vibration type, we conducted experiments measuring the tactile sensation of cloth with and without vibrations. The cloth used for the test was a piece of cotton linen (50% cotton, 50% linen) cut into 7cm×7cm pieces, and the test sample was pasted on a cardboard with adhesive tape. To maintain the surface condition of the texture, the cloth was replaced with a virgin cloth for each participant. The participants were asked to touch the cloth surface twice for each unit with only the index finger of their non-dominant hand to which the vibrator was attached, as shown in Figure 6. After touching the cloth surface for the first time, the participants were asked to remove their fingers from the cloth surface, and approximately 10 s later, they were prompted to touch it a second time. In both cases, the time required to touch the surface was set to 30 s. Whether vibration was applied to the participant's finger during the first or second touch was randomly set. After the second touch, the participants were asked to indicate whether there was a difference between the first and second sensations. As mentioned above, even if vibration was applied, the vibration intensity was below the threshold for each participant (0.7T), and the participants were unable to perceive when the vibration was applied.

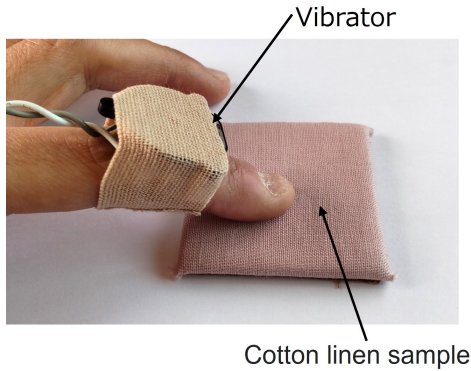


Figure 6: Illustration of the cloth tracing test.

3.3 Statistical analysis of data

During the experiments, we considered two cases: one using white noise and another using vibration at the optimal frequency for each participant. Six units were tested for each vibration case, and the first and second vibration patterns were randomly changed. We also randomly selected whether sinusoidal waves or white noise was used. Therefore, the total number of tests per participant was $2 \times 6 = 12$ units. The answer was labelled correct if the participant answered that there was a difference in the tactile sensation when the vibrations set for the first and second times were different or if they answered that there was no difference in the tactile sensation when the vibrations set for the first and second times were the same. The other answer patterns were considered incorrect. Therefore, in the protocol of this experiment, even if there was no difference owing to vibration, the correct answer rate was 33%. Because the

purpose of this experiment was to examine the difference in tactile sensation due to the difference introduced by noise, responses when no vibration was applied in both the first and second tests were excluded from the following statistics.

3.4 Results and discussion

The experimental results are shown in Figure 7. Although there was no clear statistical difference, the average value had a higher accuracy rate (56%) when the optimal vibration was used than when white noise was used (50%). In addition, a remarkable feature is that, when using the optimal sinusoidal vibration for the SR effect, the accuracy rate was distributed over a larger range compared to those obtained with under the other condition.

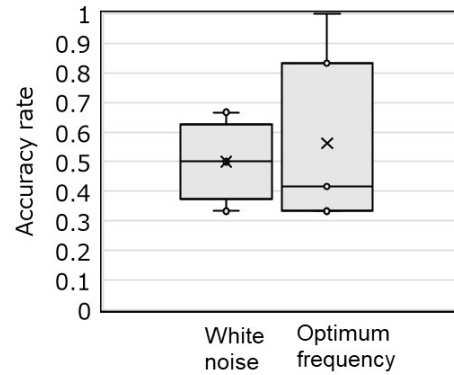


Figure 7: Correct answer rate for tests conducted using a cloth by applying different vibrations for the SR effect. Although there is no clear statistical difference, the average value exhibited a higher accuracy rate when using the optimal vibration. There were some participants whose correct answer rate increased significantly when using the optimal vibration, thus, the distribution was broadened.

To investigate the reason for the broader distribution, the distribution of the accuracy rate for each participant owing to different noise characteristics is summarized in Figure 8. Note that when using sinusoidal noise with the optimal frequency, there were some participants (namely, 3) whose accuracy rate significantly increased; therefore, the distribution was broadened, as shown in Figure 7. When these participants were interviewed after the experiment, most answered that there was a clear difference in the tactile sensation between the first and second sessions when using sinusoidal noise with the optimal frequency.

4 GENERAL DISCUSSION

Traditionally, white noise has been used as a noise source to achieve the SR effect [1–22]. However, the results reported in Section 2 show that using optimized sinusoidal vibration leads to significantly better tactile sensitivity than that obtained with white noise. This finding suggests that by finely adjusting the vibratory stimulation to match an individual's tactile sensitivity, a more personalized and comfortable user experience can be created. This optimization ensures that the stimulation remains imperceptible to the conscious

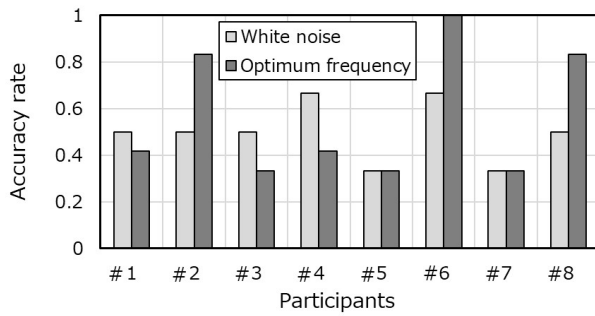


Figure 8: Changes in accuracy rate for each participant with respect to different noise types. There were some participants whose correct answer rate increased significantly when using the optimal vibration, thus, the distribution was broadened.

mind while influencing the subconscious sensory experience. In future studies, we will explore the factors related to suitable frequencies for SR.

In Section 3, we present an experiment involving a practical task using a cloth to investigate the effects of individually tuned vibrations. The test results revealed that some participants experienced a significant increase in sensory accuracy and reported clear changes in their tactile sensations. To the best of our knowledge, previous research on SR phenomena has mainly focused on statistical differences, without any documented examples of clear changes in tactile sensation. Therefore, we believe that the results of this study, which investigated using optimal-frequency noise to improve the SR effects, are significant. Although during the experiments, the correct answer rate increased when noise with the optimal frequency was used instead of white noise, the difference in the average correct answer rate was not sufficiently large to be considered statistically significant. In principle, the range of stimuli in which the SR effect is likely to occur is believed to be limited near the sensory threshold. Therefore, in addition to tests with an increased number of participants and types of cloths, a future challenge would be to explore practical the tasks during which the SR effect is more likely to occur.

In this study, the noise intensity used was 0.7 times the sensory threshold of each participant. This decision was based on previous reports indicating that adding noise with an intensity of 0.5T to 1T increases finger tactile sensitivity [1, 2, 8, 11, 14–16]. Additionally, our preliminary experiment found that tactile sensitivity increased the most at 0.5T or 0.7T, when the noise intensity was varied between 0, 0.2T, 0.5T, 0.7T, and 1T. However, note that this may not necessarily be the optimal intensity for every person. Therefore, this remains a topic for future research.

5 CONCLUSIONS

In this study, we focused on the so-called SR effect, which augments the tactile sensitivity in human fingers. We searched for conditions that would improve tactile sensitivity compared to that recorded when using conventional white noise to induce SR. The results

showed that using 200–400 Hz sinusoidal noise resulted in an expansion of finger tactile sensitivity by 28%, which is much better than 6% using white noise. The optimal frequency of the sinusoidal noise for expanding finger tactile sensitivity varies from person to person.

As an application example to investigate the SR effect obtained when using the optimal noise frequency investigated in this study, we conducted a tactile evaluation test using a cloth. This task was used to verify the effect of touching the textile twice, namely, with and without added noise. The results demonstrated that using optimal noise for each participant increased the accuracy rate. In particular, some participants exhibited a marked improvement in sensitivity.

This study reveals, for the first time, that using stochastic resonance to tailor vibration stimulation to individual abilities and needs creates new possibilities for innovation and human-centered design.

ACKNOWLEDGMENTS

I would like to thank all the participants who volunteered for the study. Moreover, I would like to extend my sincere gratitude to Dr. Urakami and Dr. Kimpara for their assistance in preparing the paper.

REFERENCES

- [1] Komi Chamnongthai, Takahiro Endo, Shohei Ikemura, and Fumitoshi Matsuno. 2020. Stiffness discrimination by two fingers with stochastic resonance. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 12272 LNCS, 497–505. https://doi.org/10.1007/978-3-030-58147-3_55
- [2] Komi Chamnongthai, Takahiro Endo, and Fumitoshi Matsuno. 2023. Two-finger Stiffness Discrimination with the Stochastic Resonance Effect. *ACM Transactions on Applied Perception* 21 (1 2023), 6:1–6:17. Issue 2. <https://doi.org/10.1145/3630254>
- [3] Komi Chamnongthai, Takahiro Endo, Fumitoshi Matsuno, Kenta Fujimoto, and Marina Kosaka. 2020. Two-Dimensional Fingertip Force Training with Improved Haptic Sensation via Stochastic Resonance. *IEEE Transactions on Human-Machine Systems* 50 (12 2020), 593–603. Issue 6. <https://doi.org/10.1109/THMS.2020.3022859>
- [4] Komi Chamnongthai, Takahiro Endo, Sajid Nisar, Fumitoshi Matsuno, Kenta Fujimoto, and Marina Kosaka. 2019. Fingertip Force Learning with Enhanced Haptic Sensation using Stochastic Resonance. *2019 IEEE World Haptics Conference, WHC 2019*, 539–544. <https://doi.org/10.1109/WHC.2019.8816114>
- [5] J J Collins, Thomas T Imhoff, and Peter Grigg. 1997. Noise-mediated enhancements and decrements in human tactile sensation. *Physical Review E* 56 (1997), 923–926. Issue 1.
- [6] James J. Collins, Attila A. Prinplata, Denise C. Gravelle, James Niemi, Jason Harry, and Lewis A. Lipsitz. 2003. Noise-Enhanced Human Sensorimotor Function. *IEEE ENGINEERING IN MEDICINE AND BIOLOGY MAGAZINE* MARCH/APRIL (2003), 76–83.
- [7] Marius Dettmer, Amir Pourmoghaddam, Beom Chan Lee, and Charles S. Layne. 2015. Effects of aging and tactile stochastic resonance on postural performance and postural control in a sensory conflict task. *Somatosensory and Motor Research* 32 (4 2015), 128–135. Issue 2. <https://doi.org/10.3109/08990220.2015.1004045>
- [8] Leah R. Enders, Pilwon Hur, Michelle J. Johnson, and Na Jin Seo. 2013. Remote vibrotactile noise improves light touch sensation in stroke survivors' fingertips via stochastic resonance. *Journal of NeuroEngineering and Rehabilitation* 10 (2013), 1–8. Issue 105. <https://doi.org/10.1186/1743-0003-10-105>
- [9] Sandeep Zechariah George, Hooman Khosravi, Ryan Peters, Lora Oehlberg, and Sonny Chan. 2019. Improving texture discrimination in virtual tasks by using stochastic resonance. *CHI'19 Extended Abstracts*, LBW0156, 1–6. <https://doi.org/10.1145/3290607.3312839>
- [10] Denise C Gravelle, Carrie A Laughton, Neel T Dhruv, Kunal D Katdare, James B Niemi, Lewis A Lipsitz, and James J Collins. 2002. Noise-enhanced balance control in older adults. *NeuroReport* 13 (2002), 1853–1856. Issue 15.
- [11] Shohei Ikemura, Takahiro Endo, and Fumitoshi Matsuno. 2021. Multiple Remote Vibrotactile Noises Improve Tactile Sensitivity of the Fingertip via Stochastic

- Resonance. *IEEE Access* 9 (2021), 17011–17019. <https://doi.org/10.1109/ACCESS.2021.3053297>
- [12] J. Collins James, T. Imhoff Thomas, and Grigg Peter. 1996. Noise-enhanced tactile sensation. *NATURE* 383 (10 1996), 770–770. Issue 31.
- [13] Mone Konno, Shigeo Yoshida, Tomohiro Amemiya, and Takuji Narumi. 2022. Improving Motivation by Increasing Tactile Sensitivity to Skin Conditions during Skin Care using Stochastic Resonance. *ACM International Conference Proceeding Series*, 317–319. <https://doi.org/10.1145/3519391.3524033>
- [14] Yuichi Kurita, Minoru Shinohara, and Jun Ueda. 2013. Wearable sensorimotor enhancer for fingertip based on stochastic resonance effect. *IEEE Transactions on Human-Machine Systems* 43 (5 2013), 333–337. Issue 3. <https://doi.org/10.1109/TSMC.2013.2242886>
- [15] Yuichi Kurita, Yamato Sueda, Takaaki Ishikawa, Minoru Hattori, Hiroyuki Sawada, Hiroyuki Egi, Hideki Ohdan, Jun Ueda, and Toshio Tsuji. 2016. Surgical Grasping Forceps with Enhanced Sensorimotor Capability via the Stochastic Resonance Effect. *IEEE/ASME Transactions on Mechatronics* 21 (12 2016), 2624–2634. Issue 6. <https://doi.org/10.1109/TMECH.2016.2591591>
- [16] Kishor Lakshminarayanan, Abigail W. Lauer, Viswanathan Ramakrishnan, John G. Webster, and Na Jin Seo. 2015. Application of vibration to wrist and hand skin affects fingertip tactile sensation. *Physiological Reports* 3 (2015), e12465. Issue 7. <https://doi.org/10.14814/phy2.12465>
- [17] Attila Priplata, James Niemi, Martin Salen, Jason Harry, Lewis A. Lipsitz, and J. J. Collins. 2002. Noise-Enhanced Human Balance Control. *Physical Review Letters* 89 (2002), 238101. Issue 23. <https://doi.org/10.1103/PhysRevLett.89.238101>
- [18] Attila A Priplata, James B Niemi, Jason D Harry, Lewis A Lipsitz, and James J Collins. 2003. Vibrating insoles and balance control in elderly people. *The LANCET* 362 (10 2003), 1123–1124. www.thelancet.com
- [19] Eri Sugimoto and Hitoshi Sasaki. 2017. Validation of a Vibrotactile Stimulation System Using the Wii Remote for Studies of Tactile Sensitivity. *The Open Psychology Journal* 10 (7 2017), 118–126. Issue 1. <https://doi.org/10.2174/1874350101710010118>
- [20] Cari Wells, Lawrence M Ward, R Chua, and J Timothy Inglis. 2005. Touch Noise Increases Vibrotactile Sensitivity in Old and Young. *PSYCHOLOGICAL SCIENCE* 16 (2005), 313–320. Issue 4.
- [21] Shun Yamashita, Kotaro Igarashi, and Naomichi Ogihara. 2021. Reducing the foot trajectory variabilities during walking through vibratory stimulation of the plantar surface of the foot. *Scientific Reports* 11 (12 2021), 7125. <https://doi.org/10.1038/s41598-021-86583-7>
- [22] Tetta Yokoyama and Yuki Hashimoto. 2021. Effect on Haptic Sensitivity of the Middle Finger of Vibrations Applied to the Little Fingernails. *2021 IEEE World Haptics Conference, WHC 2021*, 584. <https://doi.org/10.1109/WHC49131.2021.9517160>